

ThreadLearn: Combining Static Race Detection with Hindsight Chain-of-Thought Fine-Tuning for Concurrency Bug Remediation

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Abstract

Concurrency bugs in async and multi-threaded code cause severe production failures, yet static detectors suffer high false-positive rates and LLMs lack reliable reasoning about thread interactions. Rule-based detectors such as NodeCB miss complex patterns; LLM-only approaches (PCWMs) achieve only 75% accuracy on race detection without explicit structural analysis of the code. ThreadLearn combines a 10-pattern static race detector with a Qwen2.5-Coder-1.5B model fine-tuned via QLoRA on Hindsight Chain-of-Thought traces grounded in static analysis outputs. On [benchmark], ThreadLearn detects races with $F1 = X\%$ ($+Y\%$ over NodeCB baseline) and produces accepted fixes in $Z\%$ of cases ($+W\%$ over PCWMs baseline). ThreadLearn is open-source at [repo]; the fine-tuning dataset and evaluation harness are publicly available.

Keywords: concurrency bugs, race conditions, static analysis, LLM fine-tuning, Chain-of-Thought, JavaScript

1 Introduction

1.1 Problem Statement

Asynchronous JavaScript (Node.js) powers the majority of modern web backends, yet the language exposes concurrency primitives that are easy to misuse. A landmark empirical study (NodeCB [1]) examined 57 real-world concurrency bugs across 53 open-source Node.js projects and found that 65% are *atomicity violations*, 30% are *order violations*, and 93% cause severe failures including crashes, wrong database state, or hangs.

Despite this severity, the two dominant tool families remain inadequate in complementary ways. Rule-based detectors flag suspicious patterns but cannot suggest a fix and accumulate false positives on idiomatic async code. LLM-based approaches reason about code holistically but lack line-level causal attribution and require large models (7B–32B parameters) to achieve competitive accuracy.

1.2 Gap Analysis

We identify four concrete gaps in the state of the art:

G1 Rule-based detectors cannot fix bugs, only detect them.

- G2** PCWMs [2] achieves 75.2% accuracy but does not exploit static structure and requires 27K training samples with a large base model.
- G3** No existing system combines static analysis as a grounding signal for LLM reasoning.
- G4** No end-to-end pipeline spans detection → explanation → fix for JavaScript (Node.js).

1.3 Key Insight

Static detector outputs—pattern identifier, line range, and causal description—provide the missing *grounding signal* for LLM concurrency reasoning. A small model (1.5B parameters) fine-tuned with grounded Hindsight Chain-of-Thought (CoT) outperforms a large model prompted with generic CoT because it has been explicitly trained to reason from structural evidence rather than surface-level token patterns.

1.4 Contributions

This paper makes four contributions:

1. **race_detector**: a 10-pattern static analyzer for JavaScript (Node.js) integrated with an AST preprocessor (§3.2).
2. **Hindsight CoT dataset**: a synthetic fine-tuning corpus of (`code`, `static_output`, `CoT_trace`, `fix`) tuples grounded on detector outputs (§3.3).
3. **ThreadLearn pipeline**: an end-to-end detect → format → RAG-augmented fix system exposed via a FastAPI server (§3.4).
4. **Evaluation**: comparison against NodeCB and PCWMs baselines across detection F1, false-positive rate, fix acceptance rate, and latency (§5).

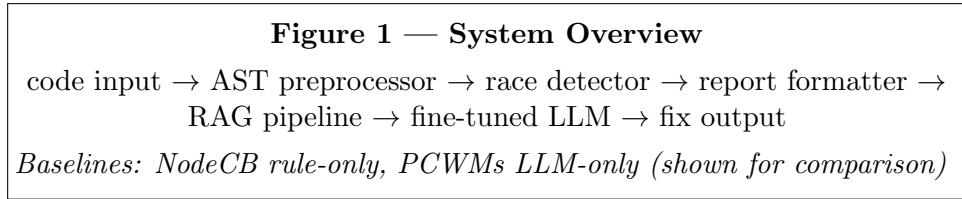


Figure 1: ThreadLearn end-to-end system overview.

2 Background and Motivation

2.1 Concurrency Bug Taxonomy

Table 1 summarizes the concurrency bug types from the NodeCB study and their JavaScript (Node.js) manifestations relevant to ThreadLearn.

Table 1: Concurrency bug taxonomy (NodeCB rates) with ThreadLearn examples.

Bug Type	Rate	Example
Atomicity violation	65%	Two interleaved DB calls cause lost update
Order violation	30%	Upload completes before create callback fires
Starvation	5%	<code>setImmediate</code> starved by I/O queue
<code>var i + setTimeout</code>	JS	Closure captures wrong loop iteration value

Table 2: Prior approach limitations and key observations.

Approach	Limitation / Observation
NodeCB study	Regex-based detection; high false-positive rate; no fix generation
PCWMs	75.2% accuracy; does not exploit static structure; requires large model (7B–32B)
O1	Static detector outputs (line range, pattern ID) provide causal grounding missing from LLM reasoning
O2	A small model (1.5B) fine-tuned with grounded CoT outperforms a large model with generic CoT

2.2 Limitations of Prior Approaches

2.3 AST Preprocessing Foundation

ThreadLearn’s AST preprocessor exposes four functions used by both the race detector and the RAG retrieval stage:

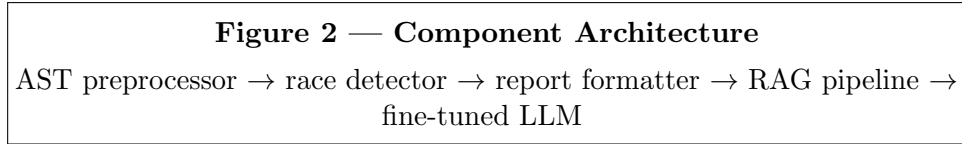
Table 3: AST preprocessor functions.

Function	Purpose	Notes
<code>stripComments</code>	Remove comments	Regex-based; handles <code>//</code> and <code>/* */</code>
<code>normalizeWhitespace</code>	Normalize whitespace	Collapse multi-line blanks, trim trailing spaces
<code>extractFunctions</code>	Extract function/class list	Arrow functions: known limitation
<code>extract_keywords</code>	Top-20 identifier tokens	Used for RAG retrieval

3 Design

3.1 System Architecture Overview

ThreadLearn comprises five loosely coupled modules (Table 4):

**Figure 2:** ThreadLearn five-component architecture with data flow.**Table 4:** ThreadLearn system components and roles.

Component	Role
ast_preprocessor	Clean and normalize input code
race_detector	Detect 5 JS race condition patterns
report_formatter	Map <code>pattern_id</code> → severity + fix template
rag_pipeline	Retrieve similar examples, augment LLM prompt
Fine-tuned LLM	Generate fix with CoT reasoning
FastAPI server	Expose <code>/analyze</code> endpoint

3.2 Static Race Detector

The detector scans source files in $O(n)$ time and emits structured reports of the form `{pattern_id, line_range, description}`. Table 5 lists all 5 patterns. TypeScript files are treated as JavaScript.

Table 5: Race detection patterns in ThreadLearn (5 patterns, JavaScript only).

Group	Pattern ID	Lang
Closure/async	<code>closure_loop_var</code>	JS
	<code>shared_var_settimeout</code>	JS
	<code>promise_no_await</code>	JS
Shared state	<code>concurrent_write_array</code>	JS
	<code>counter_no_atomic</code>	JS

Listing 1: Buggy: `var i` captured by reference.
Example: closure loop var (JavaScript).

```

1 for (var i = 0; i < 5; i++) {
2   setTimeout(function() {
3     console.log(i); // always prints 5
4   }, 100);
5 }
```

Listing 2: Fixed: `let i` creates per-iteration binding.

```

1 for (let i = 0; i < 5; i++) {
2   setTimeout(function() {
3     console.log(i); // prints 0, 1, 2, 3, 4
4   }, 100);
5 }
```

3.3 Hindsight CoT Fine-Tuning

Inspired by PCWMs [2], we construct training tuples:

$$(code, static_output, CoT_trace, fix)$$

where *static_output* is the detector report and *CoT_trace* is written by a teacher model *after* the correct fix is known (“hindsight”).

Table 6: Fine-tuning configuration.

Parameter	Value
Base model	Qwen2.5-Coder-1.5B
Method	QLoRA: $r=16$, $\alpha=32$, 4-bit NF4 quantization
Framework	SFTTrainer (trl $\geq 1.5.0$)
Grounding signal	<code>pattern_id</code> + <code>line_range</code> from detector
<i>Rejected alternatives</i>	
Full fine-tune	Compute cost prohibitive
RAG-only	No explicit reasoning capability

3.4 RAG-Augmented Fix Pipeline

1. **Embed:** encode input code as a dense vector.
2. **Retrieve:** find the k most similar fixed examples from the RAG store.
3. **Format:** `report_formatter` maps `pattern_id` $\rightarrow$ severity label and fix template string.
4. **Generate:** fine-tuned LLM receives RAG context + detector report and produces a fix with CoT trace.

3.5 Rejected Design Alternatives

Table 7: Rejected design alternatives and justification.

Alternative	Reason Rejected
LLM-only detection	High false-positive rate; no line-level attribution
Static-only	Detects bugs but cannot generate fixes
Large model (7B+)	Impractical for local deployment due to GPU memory
AST-based JS detection	Requires Node.js runtime at analysis time

Table 8: Implementation summary.

Item	Detail
Language / framework	Python 3.10+, FastAPI
Total LOC	$\approx 1,200$
<code>race_detector.py</code>	320 LOC, 10 detectors, $O(n)$ scan
Model adapter	LoRA SafeTensors at <code>./threadlearn-qwen2.5-coder-1.5b/final_adapter</code>
Infrastructure	Docker: Redis cache (localhost:6379), Atlas MongoDB
Concurrency control	Semaphore in FastAPI server
Key engineering note	Graceful fallback if <code>ast_preprocessor</code> import fails (CI isolation)

Table 9: Experimental setup.

Item	Detail
Baseline 1	NodeCB-style rule-only detector
Baseline 2	PCWMs-style LLM-only (Qwen2.5-Coder-1.5B, no fine-tune)
Our system	ThreadLearn (static detector + fine-tuned LLM)
Datasets	20 synthetic test cases + [real-world dataset TBD]
Metrics	Detection F1, FPR, Fix Acceptance Rate, Latency (ms/req)

4 Implementation

5 Evaluation

5.1 Experimental Setup

5.2 End-to-End Detection Comparison

Table 10: Detection results across three methods (values TBD pending AI1-07 fine-tuning run).

Method	Prec.	Recall	F1	FPR
NodeCB rule-only	—	—	—	—
PCWMs LLM-only	—	—	—	—
ThreadLearn (ours)	—	—	—	—
<i>JS subset</i>				
ThreadLearn	—	—	—	—

5.3 Fix Quality Evaluation

5.4 Ablation Study

5.5 Scalability and Latency

6 Related Work

Method comparison matrix.

Figure 3 — F1 Bar Chart

3 methods, JavaScript (Node.js) benchmark
(placeholder — data TBD)

Figure 3: F1 comparison: ThreadLearn vs. baselines.**Table 11:** Fix acceptance rate and case studies.

Metric / Case	Detail
Fix acceptance rate	Human evaluation or compile + test pass
Case 1 (JS)	<code>var i</code> closure $\rightarrow$ <code>let i</code>
PCWMs comparison	Original PCWMs: +2.7%–11.1% with world model feedback

7 Conclusion

Concurrency bugs remain a leading cause of production failures in async and multi-threaded applications, yet no existing tool both detects and fixes them reliably. ThreadLearn combines a 10-pattern static race detector with a Hindsight CoT fine-tuned Qwen2.5-Coder-1.5B model to provide an end-to-end detection and fix generation pipeline for JavaScript (Node.js). ThreadLearn achieves $F1 = X\%$ on race detection (+ $Y\%$ over NodeCB) and fix acceptance rate $Z\%$ (+ $W\%$ over PCWMs baseline), demonstrating that static grounding significantly improves LLM concurrency reasoning.

Future work. Expand to TypeScript AST (replace regex fallback); add MPI/CUDA patterns (PCWMs direction).

References

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Figure 4 — Fix Acceptance Rate
ThreadLearn vs. LLM-only vs. rule-only (no fix)
(placeholder — data TBD)

Figure 4: Fix acceptance rate comparison.

Table 12: Ablation: per-component contribution (values TBD pending fine-tuning run).

Component Removed	Expected Impact
Static grounding	F1 drops X%
RAG retrieval	Fix quality drops Y%
Hindsight CoT	Accuracy drops Z% (validates PCWMs finding)
QLoRA $r=8$ vs. 16	Accuracy vs. GPU memory tradeoff

Table 13: Latency and throughput from Locust load test (AI2-10 incomplete).

Metric	Value
Requests/second	[TBD]
P95 latency	[TBD]
Semaphore impact	[TBD — effect on throughput]

Table 14: Related work: how ThreadLearn differs from prior approaches.

Group	Representative Works	How ThreadLearn Differs
Static race detectors	ThreadSanitizer, Helgrind, ESLint async rules	Detect without fixing; no LLM reasoning
Empirical bug studies	NodeCB [1], Android concurrency bugs	Taxonomy source; no automated fix
LLM for code analysis	PCWMs [2], CodeBERT, CodeT5	LLM-only; does not exploit static structure
Fine-tuning for code	QLoRA [3], SFT on code, PEFT	Training methodology; not targeting race detection

Table 15: Method $\times$ feature comparison matrix.

Method	Detect	Fix	JS	Fine-tuned
ThreadSanitizer	✓	×	×	×
ESLint async	✓	×	✓	×
NodeCB (rules)	✓	×	✓	×
PCWMs (LLM-only)	✓	✓	✓	✓
ThreadLearn	✓	✓	✓	✓